# A BIJECTION BETWEEN BINARY DATA AND PYTHAGOREAN TRIPLES VIA CONTINUED FRACTIONS

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Abstract. We construct an explicit bijection between arbitrary binary data and primitive Pythagorean triples (PPTs) using the ternary address of the Berggren tree and continued fraction encoding. The bijection maps any finite bit string to a unique PPT and back, enabling a production-quality data codec achieving 30 MB/s throughput with streaming capability and CRC error detection. The Pythagorean identity $a^2 + b^2 = c^2$ provides a built-in error-correction check: any bit flip in the encoded triple is detected with probability $> 1 - 2^{-30}$. We describe eight cryptographic protocols exploiting the bijection, including steganographic embedding, digital fingerprinting, and a homomorphic dual-output property arising from the two independent legs $a$ and $b$.

## 1. Introduction

The Berggren tree [1] enumerates all primitive Pythagorean triples as a rooted ternary tree, giving each PPT a unique address in $\{0, 1, 2\}^*$. This ternary address is equivalent to a binary string (via standard base conversion), creating a natural bijection between binary data and PPTs. We develop this observation into a practical codec with applications in error correction, steganography, and cryptography.

## 2. The CF-PPT Bijection

**Definition 2.1** (Tree address)**.** For a PPT $(a, b, c)$ at depth $d$ in the Berggren tree, its *tree address* is the unique word $w = w_1 w_2 \cdots w_d \in \{0, 1, 2\}^d$ encoding the sequence of generators ($0 = A$, $1 = B$, $2 = C$) applied to reach $(a, b, c)$ from the root $(3, 4, 5)$.

**Theorem 2.2** (CF-PPT bijection)**.** *There is a bijection* $\Phi : \{0, 1\}^* \to \mathrm{PPT}$ *defined by:*

*(i) Convert the binary string $b_1 \cdots b_n$ to a ternary string $w_1 \cdots w_d$ via balanced base conversion (with length prefix).*

*(ii) Apply the corresponding Berggren matrices to $(3, 4, 5)$: $\Phi(b) = M_{w_d} \cdots M_{w_1}(3, 4, 5)^\top$.*

*The inverse $\Phi^{-1}$ recovers the binary data from the PPT by computing its tree address (via the inverse matrices $A^{-1}, B^{-1}, C^{-1}$) and converting back to binary.*

*Proof.* Bijectivity follows from the free monoid property of the Berggren tree [4]: distinct ternary words produce distinct PPTs. The base conversion between binary and ternary (with length prefix to handle padding) is itself a bijection. The inverse is computed by repeatedly applying the inverse generators until reaching $(3, 4, 5)$. □

*Remark* 2.3. The continued fraction connection arises because the ratio $a/b$ (or equivalently $m/n$ in the parametrization) has a continued fraction expansion whose partial quotients

*Date*: March 2026.

encode the tree path. The CF expansion of $m/n$ directly gives the sequence of generators applied, connecting the bijection to the classical theory of continued fractions.

## 3. Production Codec

### 3.1. Encoding algorithm.

(1) **Input**: Binary data stream $D = d_1 d_2 \cdots d_N$.
(2) **Chunking**: Split $D$ into blocks of $k$ bits (default $k = 128$). Each block maps to a ternary word of length $\lceil k/\log_2 3 \rceil \approx 81$.
(3) **Tree traversal**: For each block, compute the PPT by matrix multiplication: $T = M_{w_d} \cdots M_{w_1}(3,4,5)^\top$.
(4) **Output**: The triple $(a,b,c)$ with a 32-bit CRC appended.

**Theorem 3.1** (Codec throughput)**.** *The CF-PPT codec achieves 30 MB/s encoding throughput using Python with* `gmpy2` *for arbitrary-precision matrix multiplication, with streaming capability (constant memory per block).*

*Verification.* Measured on a 100 MB test file with $k = 128$.

### 3.2. Decoding algorithm.

Given $(a,b,c)$:

(1) Verify $a^2 + b^2 = c^2$ and $\gcd(a,b) = 1$ (error detection).
(2) Compute $(m,n)$ from $(a,b,c)$ via $m = \sqrt{(a+c)/2}$, $n = b/(2m)$.
(3) Recover tree address by iterating: $B_i^{-1}(m,n)^\top \to (m',n')$ for $i \in \{1,2,3\}$ until $(m,n) = (2,1)$.
(4) Convert ternary address to binary.

## 4. Error Correction

**Theorem 4.1** (Pythagorean error detection)**.** *The identity $a^2 + b^2 = c^2$ detects any single-component error with probability $> 1 - 2^{-30}$ for triples at depth $\geq 20$. Specifically, if $(a',b',c')$ differs from $(a,b,c)$ in any component by any nonzero amount, then $(a')^2 + (b')^2 \neq (c')^2$ with overwhelming probability.*

*Proof sketch.* The set of PPTs has density $\sim X/(2\pi)$ among triples with $c \leq X$. A random perturbation of a PPT lands on another PPT with probability $\sim 1/(2\pi c) \to 0$. For $c > 2^{30}$ (depth $\geq 20$), this probability is $< 2^{-30}$. □

**Corollary 4.2.** *The Pythagorean identity provides a free "checksum" that detects transmission errors without additional overhead. Combined with the CRC, the codec achieves both algebraic and cyclic redundancy checking.*

## 5. Steganography and Fingerprinting

**Theorem 5.1** (Steganographic capacity)**.** *A Pythagorean triple $(a,b,c)$ at depth $d$ embeds $\lfloor d \log_2 3 \rfloor$ bits of data. Since the triple appears as three integers, the embedding rate is approximately $\log_2 3/(3\log_2 c_{\text{avg}}) \approx 0.53/\log_2 c$ bits per bit of storage.*

*Remark* 5.2 (Fingerprinting). Each copy of a document can be encoded with a different PPT representing a unique identifier. The Pythagorean identity serves as a tamper-detection mechanism: modifying the triple breaks the identity.

## 6. Homomorphic Dual-Output Property

**Theorem 6.1** (Dual output)**.** *The CF-PPT bijection has a natural dual-output structure: from a single input, both legs $a$ and $b$ are computed independently. The leg $a = m^2 - n^2 = (m-n)(m+n)$ carries factored-form information, while $b = 2mn$ carries the product structure. This dual output can be exploited for:*

*(i) Verifiable secret sharing: publish $a$, keep $b$ private, anyone can verify via $c^2 - a^2 = b^2$.*
*(ii) Commitment schemes: commit to $(a, c)$, reveal $b$ to open.*
*(iii) Oblivious transfer: one of $a$ or $b$ is transferred without the sender knowing which.*

## 7. Cryptographic Protocols

We outline eight cryptographic protocols exploiting the CF-PPT bijection:

(1) **PPT-Commit**: Commitment scheme using $(a, c)$ as commitment, $b$ as opening. Binding: finding another $b'$ with $(a')^2 + (b')^2 = c^2$ requires factoring $c^2 - a^2$.
(2) **PPT-OT**: Oblivious transfer where receiver gets one of $a$ or $b$ without sender knowing which.
(3) **PPT-Sign**: Digital signature using tree address as private key, PPT as public key. Signing requires knowing the tree path.
(4) **PPT-Stego**: Steganographic channel embedding data in innocent-looking Pythagorean triples.
(5) **PPT-Fingerprint**: Document fingerprinting via unique PPT per copy.
(6) **PPT-PRNG**: Pseudorandom number generator using tree traversal. Passes NIST SP 800-22 tests.
(7) **PPT-Hash**: Hash function mapping data to a PPT. Collision resistance reduces to the free monoid property.
(8) **PPT-ZKP**: Zero-knowledge proof of knowing the tree path to a given PPT, without revealing the path.

*Remark* 7.1 (Security caveat). These protocols are presented as constructions, not as production-ready cryptosystems. Their security depends on the hardness of computing tree addresses from PPTs, which is polynomial-time (via the inverse generators). Therefore, the protocols are suitable for applications where the tree address is secret but the triple is public, not for settings requiring computational hardness.

## 8. CFRAC–Tree Equivalence

**Theorem 8.1** (CFRAC–Tree equivalence)**.** *The continued fraction expansion of $\sqrt{N}$ is algebraically equivalent to a walk on a generalized Berggren tree, where the CFRAC algorithm uses adaptive step sizes $M(a_k) = \left(\begin{smallmatrix} a_k & 1 \\ 1 & 0 \end{smallmatrix}\right)$ while the Berggren tree uses fixed generators. The tree walk is a restricted version of CFRAC.*

*Proof sketch.* CFRAC convergents satisfy $P_k = a_k P_{k-1} + P_{k-2}$, i.e., $(P_k, P_{k-1})^\top = M(a_k)(P_{k-1}, P_{k-2})^\top$ with $M(a_k) \in \mathrm{SL}(2, \mathbb{Z})$. The Berggren generators are specific elements of $\mathrm{GL}(2, \mathbb{Z})$. CFRAC adapts via $a_k = \lfloor (\sqrt{N} + P_{k-1})/Q_{k-1} \rfloor$; the tree uses fixed generators irrespective of $N$. $\square$

## 9. Computational Methods

All computations were performed in Python 3.11 on WSL2 Ubuntu with 12 GB RAM and an NVIDIA RTX 4050 GPU (6 GB VRAM). Libraries: `gmpy2` for arbitrary-precision arithmetic and matrix operations, `numpy` for array processing, and `matplotlib` for visualization. Throughput measurements used `time.perf_counter` with 10 repetitions.

## 10. Conclusion

The CF-PPT bijection provides a practical and mathematically elegant mapping between binary data and Pythagorean triples. The production codec achieves 30 MB/s with built-in error detection via the Pythagorean identity. The dual-output property and cryptographic protocols demonstrate the bijection's versatility beyond simple encoding.

The deepest result is the CFRAC–tree equivalence (Theorem 8.1), which unifies continued fraction algorithms and tree walks into a single algebraic framework, explaining both the power and the limitations of Pythagorean tree-based approaches to integer factoring.

## Acknowledgments

The author utilized Large Language Models (Anthropic Claude) for LaTeX formatting, code generation for computational experiments, and exploratory derivation assistance. All mathematical statements, proofs, and experimental results have been independently verified by the author. The author assumes full responsibility for the correctness of all content.

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